# 3D-LINES

## SYNOPSIS

### Equation of a line:

➤ **General Form (Unsymmetrical form) of a line:**

The intersection of two plane s is a line.The equations

$a_1x+b_1y+c_1z+d_1=0=a_2x+b_2y+c_2z+d_2$

represents a line.

i)Equation to the X-axis is y = 0 , z = 0

ii)Equation to the Y- axis is x = 0 , z = 0

iii) Equation to the Z- axis is x = 0 , y = 0

iv) Equation of the line parallel to x-axis is y=p, z=q, p,q, $\in$ R

v) Equation of the line parallel to y-axis is x=h, z=q, h, q $\in$ R

vi) Equation of the line parallel to z-axis is x=h, y=p, h, p $\in$ R

**Symmetrical form of a line:**

➤ The equation of the line passing through the point $(x_1, y_1, z_1)$ and having d.c's $(l, m, n)$ is

$$\frac{x-x_1}{l}=\frac{y-y_1}{m}=\frac{z-z_1}{n}$$

**Proof:** Let L be the line passing through $(x_1, y_1, z_1)$ with d.c's $(l, m, n)$, Let A = $(x_1, y_1, z_1)$

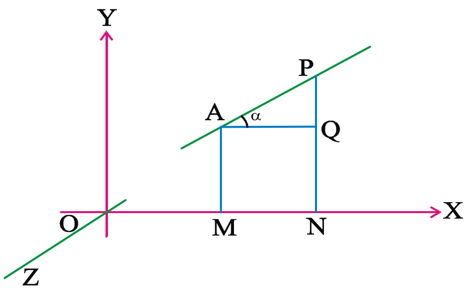
Let P(x, y, z) be any point on L Let AP = r

Let $\alpha$ be the angle which L makes with x-axis

Let M, N be the projections of A, P on x-axis respectively.

Let Q be the projections of A on $\overline{PN}$ ,

OM = $x_1$, ON = x, MN = ON – OM = x – $x_1$




Also MN = projection of AP on x-axis

$MN = r.\cos\alpha$

$x - x_1 = r.l$

$\frac{x-x_1}{l} = r$

$$\left[\begin{array}{c} From\ \Delta APQ \\ \cos\alpha = \frac{AQ}{AP} \\ AQ = AP.\cos\alpha \\ MN = r.\cos\alpha \end{array}\right]$$

Similarly we can prove $\frac{y-y_1}{m}=\frac{z-z_1}{n}=r$

Hence the equation of the line L is

$$\frac{x-x_1}{l}=\frac{y-y_1}{m}=\frac{z-z_1}{n}$$

i) The equation of a line passing through the point $(x_1, y_1, z_1)$ and having d.r's (a, b, c) is

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$$

ii) The equation of the line passing through two points $(x_1, y_1, z_1)$ and $(x_2, y_2, z_2)$ is

$$\frac{x-x_1}{x_2-x_1}=\frac{y-y_1}{y_2-y_1}=\frac{z-z_1}{z_2-z_1}$$

**Vector form of a line:**

➤ Cartesian equation of a line passing through the point $(x_1,y_1,z_1)$ and having d.r's (a,b,c) is

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$$

Let $\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}=\lambda$

$x-x_1=a\lambda \Rightarrow x=x_1+a\lambda$

$y-y_1=b\lambda \Rightarrow y=y_1+b\lambda$

$z-z_1=c\lambda \Rightarrow z=z_1+c\lambda$, Now,

$x\bar{i}+y\bar{j}+z\bar{k}=x_1\bar{i}+\lambda a\bar{i}+y_1\bar{j}+\lambda b\bar{j}+z_1\bar{k}+\lambda c\bar{k}$

$\bar{r}=(x_1\bar{i}+y_1\bar{j}+z_1\bar{k})+\lambda(a\bar{i}+b\bar{j}+c\bar{k})$

Which is the vector equation of the line passing through $(x_1,y_1,z_1)$ and having d.r's (a,b,c) (or) vector equation of the line passing through $(x_1, y_1, z_1)$ and parallel to the vector $a\bar{i} + b\bar{j} + c\bar{k}$ where $\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$.

***W.E-1:** The cartesian equation and vector equation of the line passing through the points (–1, 0, 2) and (3, 4, 6)*

**Sol:** Let $(x_1, y_1, z_1) = (-1, 0, 2)$

$(x_2, y_2, z_2) = (3, 4, 6)$

D.r's of the line $= (x_2 - x_1, y_2 - y_1, z_2 - z_1)$

let $(a,b,c)=(4, 4, 4)$

Cartesian equation of the line is

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

$$\Rightarrow \frac{x+1}{4} = \frac{y-0}{4} = \frac{z-2}{4}$$

Let $\frac{x+1}{4} = \frac{y-0}{4} = \frac{z-2}{4} = \lambda$

$x + 1 = 4\lambda \Rightarrow x = -1 + 4\lambda$

$y = 4\lambda$

$z - 2 = 4\lambda \Rightarrow z = 2 + 4\lambda$

$x\bar{i} + y\bar{j} + Z\bar{k} = -\bar{i} + 4\lambda\bar{i} + 4\lambda\bar{j} + 2\bar{k} + 4\lambda\bar{k}$

$\bar{r} = (-\bar{i} + 2\bar{k}) + \lambda(4\bar{i} + 4\bar{k})$

Where $\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$

Which is the vector equation of the line.

### Conversion of non-symmetrical form to symmetrical form:

➢ Let the equation of the line in non-symmetrical form be

$a_1x + b_1y + c_1z + d_1 = 0 = a_2x + b_2y + c_2z + d_2$

To find the equation of the line in symmetrical form, we must know (i) its d.r's (ii) coordinates of any point on it.

**i) To find the d.rs of the line:**

Let $l$, m, n be the d.r's of the line.

Since the line lies in both the planes, it must be perpendicular to normals of both planes.

So, $a_1l + b_1m + c_1n = 0$

$a_2l + b_2m + c_2n = 0$

From these equations proportional values of $l, m, n$ found by cross multiplication method

$$\begin{array}{cccc} & l & m & n \\ b_1 & c_1 & a_1 & b_1 \\ b_2 & c_2 & a_2 & b_2 \end{array}$$

$$\Rightarrow \frac{l}{b_1c_2 - b_2c_1} = \frac{m}{c_1a_2 - c_2a_1} = \frac{n}{a_1b_2 - a_2b_1}$$

**ii) To find a point on the line:**

At least one of the d.r's must be non-zero.

Let $a_1b_2 - a_2b_1 \neq 0$

The line cannot be parallel to xy-plane.

Let it intersect the xy-plane in $(x_1, y_1, 0)$

then $a_1x_1 + b_1y_1 + d_1 = 0$

and $a_2x_1 + b_2y_1 + d_2 = 0$

By solving these equations we get the point $(x_1,y_1,0)$ on the line.

Hence the equation of the line in symmetric form is $\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z-0}{n}$

**Note:** If $l \neq 0$, take a point on yz-plane as $(0,y_1,z_1)$ and if $m \neq 0$ take a point on xz-plane as $(x_1,0,z_1)$.

***W.E-2:** The equation of the line $3x + 2y - z - 4 = 0 = 4x + y - 2z + 3$ in symmetrical form is*

**Sol:** Let (a, b, c) be the d.r's of the line then

$3a + 2b - c = 0,\ 4a + b - 2c = 0$

By cross multiplication method

$$\frac{a}{-3} = \frac{b}{2} = \frac{c}{-5}$$

d.r's of the line are (–3, 2, –5)

Since $c \neq 0$, the line not parallel to xy-plane.

Let $(x_1, y_1, 0)$ be a point on the line

$3x_1 + 2y_1 - 4 = 0,\ 4x_1 + y_1 + 3 = 0$

By solving these equations we get

$x_1 = -2, y_1 = 5$

A point on the line is (–2, 5, 0)

Hence the equation of the line in symmetrical form

$$\frac{x+2}{-3} = \frac{y-5}{2} = \frac{z-0}{-5}$$

### Parametric form:

➢ The parametric equations of the line passing through the point $P(x_1, y_1, z_1)$ and having d.c's $(l, m, n)$ are $x = x_1 + lr$, $y = y_1 + mr$, $z = z_1 + nr$ Where r = OP

**Remark:** The coordinates of a point on the line whose d.c's are ($l$, m, n) which is at a distance of 'r' units from the point $(x_1, y_1, z_1)$ are $(x_1 \pm lr, y_1 \pm mr, z_1 \pm nr)$

**W.E-3.** ***The coordinates of a point on the line $\frac{x-1}{2} = \frac{y+1}{-3} = z$ at a distance of $4\sqrt{14}$ from the point (1, –1, 0) nearer to the origin are***

**Sol.** Let A = (1,–1,0)

$\frac{x-1}{2} = \frac{y+1}{-3} = z = t$

The cooridnates of any point P on the given line are (2t + 1, –3t – 1, t)

$AP = 4\sqrt{14}$

$(2t)^2 + (-3t)^2 + (t)^2 = (4\sqrt{14})^2 \Rightarrow t = \pm 4$

So the coordinates of the required point are (9, –13, 4) and (–7, 11, –4)

Out of which nearer to the origin is (–7, 11, –4)

### Angle between two lines:

➢ If $\theta$ is the angle between the lines given by

$\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$ and

$\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$ then

$$\cos\theta = \left| \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{\sum a_1^2}\sqrt{\sum a_2^2}} \right|$$

i)a)If the lines are parellel then $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

b) If the lines are $\perp^r$ then $a_1a_2 + b_1b_2 + c_1c_2 = 0$

ii) If '$\theta$' is the acute angle between the line $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ and the plane $ax + by + cz + d = 0$ then

$$\sin\theta = \frac{|al + bm + cn|}{\sqrt{a^2 + b^2 + c^2}.\sqrt{l^2 + m^2 + n^2}}$$

iii) If the line $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ is parallel to the plane $ax + by + cz + d = 0$ then $al + bm + cn = 0$ (Normal to the plane is perpendicular to the line)

iv) If the line $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ perpendicular to the plane $ax + by + cz + d = 0$ then $\frac{a}{l} = \frac{b}{m} = \frac{c}{n}$.

v) d.c's of the line which makes equal angles with coordinate axes are $\pm\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ and the d.r's of the line are (1, 1, 1).

### Coplanar lines:

➢ Two lines are said to be coplanar if they are either parallel or intersect.

### Non–Coplanar Lines:

➢ Two lines are said to be non coplanar or skew lines if they are neither parallel nor intersecting.

### Condition for two lines to be coplanar:

➢ The line $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ lies in the plane $ax + by + cz + d = 0$ if

$ax_1 + by_1 + cz_1 + d = 0$, $al + bm + cn = 0$.

**Proof:** *let* $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$.............(1)

$ax + by + cz + d = 0$..........(2)

if the line (1) lies in the plane (2) then for all values '$t$' the point $p(x_1 + lt, y_1 + mt, z_1 + nt)$ will lie in the plane (2)

ie, $a(x_1 + lt) + b(y_1 + mt) + c(z_1 + nt) + d = 0$

$\Rightarrow t(al+bm+cn)+(ax_1+by_1+cz_1+d)=0$

it is true for all values of 't'

hence we must have $al+bm+cn=0$ and $ax_1+by_1+cz_1+d=0$

which are the required conditions.

**Remark:** The lines $\frac{x-x_1}{a_1}=\frac{y-y_1}{b_1}=\frac{z-z_1}{c_1}$, $\frac{x-x_2}{a_2}=\frac{y-y_2}{b_2}=\frac{z-z_2}{c_2}$ are coplanar

$$\Leftrightarrow \begin{vmatrix} x_1-x_2 & y_1-y_2 & z_1-z_2 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}=0$$

***W.E-4.*** ***The lines*** $\frac{x-a+d}{\alpha-\delta}=\frac{y-a}{\alpha}=\frac{z-a-d}{\alpha+\delta}$ ***and*** $\frac{x-b+c}{\beta-\gamma}=\frac{y-b}{\beta}=\frac{z-b-c}{\beta+\gamma}$ ***are***

**Sol:** $(x_1,y_1,z_1)=(a-d,a,a+d)$

$(x_2,y_2,z_2)=(b-c,b,b+c)$

$(a_1,b_1,c_1)=(\alpha-\delta,\alpha,\alpha+\delta)$

$(a_2,b_2,c_2)=(\beta-\gamma,\beta,\beta+\gamma)$

now $\begin{vmatrix} x_1-x_2 & y_1-y_2 & z_1-z_2 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}$

$$=\begin{vmatrix} a-d-b+c & a-b & a+d-b-c \\ \alpha-\delta & \alpha & \alpha+\delta \\ \beta-\gamma & \beta & \beta+\gamma \end{vmatrix}$$

$c_1 \to c_1+c_2+c_3$

$$=\begin{vmatrix} 3(a-b) & a-b & a+d-b-c \\ 3\alpha & \alpha & \alpha+\delta \\ 3\beta & \beta & \beta+\gamma \end{vmatrix}=0$$

Hence given lines are coplanar

## . Equation of a plane containing lines:

➤ The equation of the plane containing the lines

$$\frac{x-x_1}{a_1}=\frac{y-y_1}{b_1}=\frac{z-z_1}{c_1},\ \frac{x-x_2}{a_2}=\frac{y-y_2}{b_2}=\frac{z-z_2}{c_2}$$

is $\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}=0$ (or)

$$\begin{vmatrix} x-x_2 & y-y_2 & z-z_2 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}=0$$

***W.E-5*** ***The value of K, if the lines*** $\frac{x+3}{-3}=\frac{y-1}{1}=\frac{z-k}{5}$ ***and*** $\frac{x+1}{-1}=\frac{y-2}{2}=\frac{z-5}{5}$ ***are coplanar***

**Sol:** points on the given lines $(x_1,y_1,z_1)=(-3,1,k)$

$(x_2,y_2,z_2)=(-1,2,5)$

D.r's of given lines $(a_1,b_1,c_1)=(-3,1,5)$

$(a_2,b_2,c_2)=(-1,2,5)$

Equation of the plane is

$$\begin{vmatrix} x_1-x_2 & y_1-y_2 & z_1-z_2 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}=0$$

$$\Rightarrow \begin{vmatrix} -2 & -1 & k-5 \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix}=0 \Rightarrow K=5$$

➤. If the lines $\frac{x-x_1}{l}=\frac{y-y_1}{m}=\frac{z-z_1}{n}$,

$a_1x+b_1y+c_1z+d_1=0=a_2x+b_2y+c_2z+d_2$ are coplanar then

$$\frac{a_1x_1+b_1y_1+c_1z_1+d_1}{a_1l+b_1m+c_1n}=\frac{a_2x_1+b_2y_1+c_2z_1+d_2}{a_2l+b_2m+c_2n}$$

## Skew lines:

➢ Two straight lines are said to be skew lines if they are neither parallel nor intersecting. i.e. the lines which do not lie in a plane.

### Shortest distance:

➢ If $L_1$ and $L_2$ are skew lines then there is one and only one line perpendicular to both of the lines $L_1$ and $L_2$ which is called the line of shortest distance. If PQ is the line of shortest distance then the distance between P and Q is called distance between the given skew lines.

i) The shortest distance between the skew lines $\bar{r}=\bar{a}_1+\lambda\bar{b}_1, \bar{r}=\bar{a}_2+\mu\bar{b}_2$ is

$$\frac{|(\bar{a}_1-\bar{a}_2)\cdot(\bar{b}_1\times\bar{b}_2)|}{|\bar{b}_1\times\bar{b}_2|}\ (or)\ \frac{\left|\left[\bar{a}_1-\bar{a}_2\ \ \bar{b}_1\ \ \bar{b}_2\right]\right|}{|\bar{b}_1\times\bar{b}_2|}$$

ii) If the above two lines are coplanar or intersecting then $\left[\bar{a}_1-\bar{a}_2\ \ \bar{b}_1\ \ \bar{b}_2\right]=0$

iii) Shortest distance between the lines

$$\frac{x-x_1}{a_1}=\frac{y-y_1}{b_1}=\frac{z-z_1}{c_1}$$

and $\dfrac{x-x_2}{a_2}=\dfrac{y-y_2}{b_2}=\dfrac{z-z_2}{c_2}$ is

$$\left.\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}\right/\sqrt{\sum(b_1c_2-b_2c_1)^2}$$

**W.E-6:** ***The shortest distance beween the lines whose vector equations are***

$\bar{r}=\left(\bar{i}+\bar{j}\right)+\lambda(2\bar{i}-\bar{j}+\bar{k})$ ***and***

$\bar{r}=\left(2\bar{i}+\bar{j}-\bar{k}\right)+\mu(3\bar{i}-5\bar{j}+2\bar{k})$ ***is***

**Sol:** Compare the given equations with $\bar{r}=\bar{a}_1+\lambda\bar{b}_1, \bar{r}=\bar{a}_2+\mu\bar{b}_2$

$\therefore \bar{a}_1=\bar{i}+\bar{j}, \bar{b}_1=2\bar{i}-\bar{j}+\bar{k}$

$\bar{a}_2=2\bar{i}+\bar{j}-\bar{k}, \bar{b}_2=3\bar{i}-5\bar{j}+2\bar{k}$

$\therefore \bar{a}_1-\bar{a}_2=-\bar{i}+\bar{k}$

$$\bar{b}_1\times\bar{b}_2=\begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 2 & -1 & 1 \\ 3 & -5 & 2 \end{vmatrix}=3\bar{i}-\bar{j}-7\bar{k}$$

$$\left|\bar{b}_1\times\bar{b}_2\right|=\sqrt{9+1+49}=\sqrt{59}$$

$$\left[\bar{a}_1-\bar{a}_2\ \ \bar{b}_1\ \ \bar{b}_2\right]=\begin{vmatrix} -1 & 0 & 1 \\ 2 & -1 & 1 \\ 3 & -5 & 2 \end{vmatrix}=-10$$

Shortest distance between the skew lines

$$=\frac{\left|\left[\bar{a}_1-\bar{a}_2\ \ \bar{b}_1\ \ \bar{b}_2\right]\right|}{\left|\bar{b}_1\times\bar{b}_2\right|}=\frac{10}{\sqrt{59}}$$

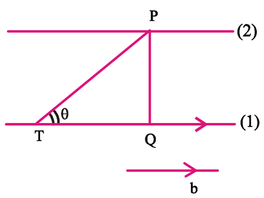
### Distance between parallel lines:

➢ The distance between the parallel lines $\bar{r}=\bar{a}_1+\lambda\bar{b}, \bar{r}=\bar{a}_2+\mu\bar{b}$ is $\dfrac{\left|\bar{b}\times(\bar{a}_1-\bar{a}_2)\right|}{|\bar{b}|}$

**Proof:** Given parallel lines are:

$\bar{r}=\bar{a}_1+\lambda\bar{b}$ --- (1)

$\bar{r}=\bar{a}_2+\lambda\bar{b}$ --- (2)




Let PQ be the distance between (1) and (2)

let T be a point on (1) with $OT=\bar{a}_1$

Let $\overline{OP}=\bar{a}_2$

Let Q be the projection of P on (1)

Let $\theta$ be the angle between PT and $\bar{b}$

$\bar{b}\times\overline{TP}=\left(|\bar{b}||\overline{TP}|\sin\theta\right).\hat{n}$ --- (3)

Where $\hat{n}$ is the unit vector perpendicular to the plane of the lines (1) and (2)

$\overline{TP}=\overline{OP}-\overline{OT}=\bar{a}_2-\bar{a}_1$

$$\left[\begin{array}{l} In\ \Delta PTQ \\ \sin\theta = \frac{PQ}{PT} \Rightarrow PQ = PT.\sin\theta \end{array}\right]$$

From (3) $\quad \bar{b}\times\overline{TP} = \left(|\bar{b}|\,|\overline{TP}|\sin\theta\right)\hat{n}$

$$\Rightarrow \bar{b}\times(\bar{a}_2-\bar{a}_1) = |\bar{b}|\,(PQ)\hat{n}$$

$$\Rightarrow |\bar{b}\times(\bar{a}_2-\bar{a}_1)| = |\bar{b}|\cdot|\overline{PQ}| \qquad (|\hat{n}|=1)$$

$$\Rightarrow |\overline{PQ}| = \frac{|\bar{b}\times(\bar{a}_2-\bar{a}_1)|}{|\bar{b}|} \Rightarrow PQ = \frac{|\bar{b}\times(\bar{a}_2-\bar{a}_1)|}{|\bar{b}|}$$

$$\Rightarrow PQ = \frac{|\bar{b}\times(\bar{a}_1-\bar{a}_2)|}{|\bar{b}|}$$

**W.E-7** ***The distance between the parallel lines given by*** $\bar{r} = (\bar{i}+2\bar{j}-4\bar{k}) + \lambda(2\bar{i}+3\bar{j}+6\bar{k})$ ***and*** $\bar{r} = (3\bar{i}+3\bar{j}-5\bar{k}) + \mu(2\bar{i}+3\bar{j}+6\bar{k})$

**Sol:** Compare the given lines with $\bar{r} = \bar{a}_1 + \lambda\bar{b}$

$$\bar{r} = \bar{a}_2 + \mu\bar{b}$$

$\bar{a}_1 = \bar{i}+2\bar{j}-4\bar{k}$,

$\bar{a}_2 = 3\bar{i}+3\bar{j}-5\bar{k}$, $\quad \bar{b} = 2\bar{i}+3\bar{j}+6\bar{k}$

Distance between the lines $= \dfrac{|\bar{b}\times(\bar{a}_1-\bar{a}_2)|}{|\bar{b}|}$

$\bar{a}_1-\bar{a}_2 = -2\bar{i}-\bar{j}+\bar{k}$

$$\bar{b}\times(\bar{a}_1-\bar{a}_2) = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 2 & 3 & 6 \\ -2 & -1 & 1 \end{vmatrix} = 9\bar{i}-14\bar{j}+4\bar{k}$$

$|\bar{b}\times(\bar{a}_1-\bar{a}_2)| = \sqrt{81+196+16} = \sqrt{293}$

$|\bar{b}| = \sqrt{4+9+36} = 7$

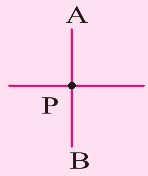
Distance between the lines $= \dfrac{\sqrt{293}}{7}$

**W.E-8** ***The image of the point (1, 6, 3) in the line*** $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ ***is***

**Sol.** Let A = (1, 6, 3)

Let P be the foot of the perpendicular from A(1, 6, 3) to the given line

$\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3} = t$




then P = (t, 2t+1, 3t+2)

D.r's of $\overline{AP}$ (t – 1, 2t – 5, 3t – 1)

$\overline{AP}$ is perpendicular to the given line

$\Rightarrow 1(t-1)+2(2t-5)+3(3t-1)=0$

$\Rightarrow t=1$, P = (1, 3, 5)

Let B be the image of A

B = 2P – A = (1, 0, 7)

## C.U.Q.

1. **The line** $m(x-a) = l(z-b)$, $y = c$ **is perpendicular to**
   1) X- axis 2) Y-axis 3) Z-axis 4) XY-plane
2. **If the lines** $x = az+b$, $y = cz+d$; $x = a_1z+b_1$, $y = c_1z+d_1$ **are perpendicular then**
   1) $aa_1+cc_1+1=0$ 2) $aa_1+cc_1-1=0$
   3) $aa_1+bb_1+1=0$ 4) $aa_1+dd_1-1=0$
3. **If the equation of line is** $x = ay+b$; $z = cy+d$**. Then its symmetric form is**
   1) $\frac{x-a}{b} = \frac{y}{1} = \frac{z-c}{d}$ 2) $\frac{x-b}{a} = \frac{y}{1} = \frac{z-d}{c}$
   3) $\frac{x+a}{b} = \frac{y}{1} = \frac{z+c}{d}$ 4) $\frac{x+b}{a} = \frac{y}{1} = \frac{z+d}{c}$

**4. Equation of a line through $(\alpha,\beta,\gamma)$ and parallel to Y-axis are**

1) $\frac{x-\alpha}{1}=\frac{y-\beta}{0}=\frac{z-\gamma}{0}$ 2) $\frac{x-\alpha}{0}=\frac{y-\beta}{1}=\frac{z-\gamma}{0}$

3) $\frac{x-\alpha}{0}=\frac{y-\beta}{0}=\frac{z-\gamma}{1}$ 4) $\frac{x-\alpha}{1}=\frac{y-\beta}{1}=\frac{z-\gamma}{1}$

**5. The equation of the plane containing the line**

$\frac{x-x_1}{l}=\frac{y-y_1}{m}=\frac{z-z_2}{n}$ **is**

**a** $(x-x_1)+b(y-y_1)+c(z-z_1)=0$ **where**

1) $ax_1+by_1+cz_1=0$ 2) $al+bm+cn=0$

3) $\frac{a}{l}=\frac{b}{m}=\frac{c}{n}$ 4) $lx_1+my_1+nz_1=0$

**6. Equation of a line through (a, b, c) and parallel to z-axis is**

1) $\frac{x-a}{1}=\frac{y-b}{0}=\frac{z-c}{0}$ 2) $\frac{x-a}{0}=\frac{y-b}{0}=\frac{z-c}{1}$

3) $\frac{x-a}{1}=\frac{y-b}{1}=\frac{z-c}{1}$ 4) $\frac{x-a}{0}=\frac{y-b}{1}=\frac{z-c}{0}$

## C.U.Q - KEY

1) 2 2) 1 3) 2 4) 2 5) 2 6) 2

## C.U.Q - HINTS

1. Find dr's of line of intersection of the given planes.
2. $\frac{x-b}{a}=\frac{y-d}{c}=\frac{z}{1}, \frac{x-b_1}{a_1}=\frac{y-d_1}{c_1}=\frac{z}{1}$
3. Write in symmetric form
4. drs of y-axis is (0, 1, 0)
5. Plane contains the given line normal to the plane must be perpendicular to the line so, $al + bm + cn = 0$.
6. d.c's of z-axis are (0, 0, 1)

## LEVEL-I - (C.W)

### EQUATION OF A LINE

**1. The equation of the line through (3,1,2) and equally inclined to the axes is**

1) $\frac{x-3}{1}=\frac{y-1}{0}=\frac{z-2}{0}$ 2) $\frac{x-3}{0}=\frac{y-1}{1}=\frac{z-2}{0}$

3) $\frac{x-3}{0}=\frac{y-1}{0}=\frac{z-2}{1}$ 4) $\frac{x-3}{1}=\frac{y-1}{1}=\frac{z-2}{1}$

**2. Equation of a line which is parallel to the line $\frac{x-2}{3}=\frac{y+1}{1}=\frac{z-7}{9}$, which passes through the point ( 3,0,5 ) is**

1) $\frac{x-2}{3}=\frac{y+1}{0}=\frac{z-7}{5}$ 2) $\frac{x-2}{0}=\frac{y+1}{1}=\frac{z-7}{4}$

3) $\frac{x-3}{3}=\frac{y-0}{1}=\frac{z-5}{9}$ 4) $\frac{x-3}{0}=\frac{y-0}{1}=\frac{z-5}{9}$

**3. The equation of the line passing through (-1, 2, -3 ) and perpdendicular to the plane** $2x+3y+z+5=0$ **is**

1) $\frac{x-1}{-1}=\frac{y+2}{1}=\frac{z-3}{-1}$ 2) $\frac{x+1}{-1}=\frac{y-2}{1}=\frac{z+3}{1}$

3) $\frac{x+1}{2}=\frac{y-2}{3}=\frac{z+3}{1}$ 4) $\frac{x+1}{1}=\frac{y-2}{1}=\frac{z+3}{1}$

**4. The cartesian equation of line is $\frac{x-5}{3}=\frac{y+4}{7}=\frac{z-6}{2}$ its vector form is**

1) $\bar{r}=(5\bar{i}+4\bar{j}+6\bar{k})+\lambda(3\bar{i}+2\bar{j}+2\bar{k})$

2) $\bar{r}=(5\bar{i}-4\bar{j}+6\bar{k})+\lambda(3\bar{i}+7\bar{j}+2\bar{k})$

3) $\bar{r}=(5\bar{i}+4\bar{j}+6\bar{k})+\lambda(3\bar{i}+7\bar{j}+2\bar{k})$

4) $\bar{r}=(-5\bar{i}+4\bar{j}+6\bar{k})+\lambda(3\bar{i}+7\bar{j}+2\bar{k})$

### ANGLE BETWEEN TWO LINES

**5. The value of p so that the lines $\frac{1-x}{3}=\frac{7y-14}{2p}=\frac{z-3}{2}$ and $\frac{7-7x}{3p}=\frac{y-5}{1}=\frac{6-z}{5}$ are at right angles are**

1) 70/11 2) 7/11 3) 10/7 4) 17/11

**6. The angle between the lines**

$\frac{x-1}{2}=\frac{y-2}{1}=\frac{z+3}{2}$ **and** $\frac{x}{1}=\frac{y}{1}=\frac{z}{0}$ **is**

1) $0^0$ 2) $30^0$ 3) $45^0$ 4) $90^0$

7. **The sine of the angle between the straight line** $\frac{x-2}{3}=\frac{y-3}{4}=\frac{z-4}{5}$ **and the plane** $2x-2y+z=5$ **is**

1) $\frac{\sqrt{5}}{3}$ 2) $\frac{2\sqrt{2}}{5}$ 3) $\frac{1}{5\sqrt{2}}$ 4) $\frac{2\sqrt{3}}{5}$

8. **The angle between the lines 2x = 3y = –z and 6x = –y = – 4z is**

1) $0^o$ 2) $30^o$ 3) $45^o$ 4) $90^o$

9. **The line passing through the points (5, 1, a) and (3, b, 1) crosses the yz-plane at the point** $\left(0, \frac{17}{2}, \frac{-13}{2}\right)$ **then (AIEEE - 2008)**

1) a = 4, b = 6 2) a = 6, b = 4
3) a = 8, b = 2 4) a = 2, b = 8

10. **The angle between the lines x = 1, y = 2 and y = –1, z = 0 is**

1) $\frac{\pi}{2}$ 2) $\frac{\pi}{6}$ 3) $\frac{\pi}{3}$ 4) $0^o$

## COPLANAR LINES

11. **The lines**

$\frac{x-1}{a}=\frac{y-2}{3}=\frac{z-3}{4}$ ; $\frac{x-2}{3}=\frac{y-3}{4}=\frac{z-4}{5}$

**are coplanar. Then a =**

1) 1 2) 2 3) -1 4) -2

12. **The line** $\frac{x+3}{3}=\frac{y-2}{-2}=\frac{z+1}{1}$ **and the plane** $4x+5y+3z-5=0$ **intersect at a point**

1) (3,1,-2) 2) (3,-2,1)
3) (2,-1,3) 4) (-1,-2,-3)

13. **If the lines** $\frac{x-2}{1}=\frac{y-3}{1}=\frac{z-4}{-k}$ **and** $\frac{x-1}{k}=\frac{y-4}{2}=\frac{z-5}{1}$ **are coplanar then k can have (MAIN-2013)**

1) any value 2) exactly one value
3) exactly two values 4) exactly three values

## INTERSECTION OF LINES

14. **A line with d.c's proportional to (2,1,2) meets each of the lines** $x=y+a=z$ **and** $x+a=2y=2z$ **. Then the coordinates of each of the points of intersection are given by**

1) (3a, 2a, 3a) ; (a, a, 2a)
2) (3a, 2a, 3a) ; ( a, a, a)
3) (3a, 3a, 3a) ; ( a, a, a)
4) (2a, 3a, 3a) ; ( 2a, a, a)

15. **If the line** $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-1}{4}$ **and** $\frac{x-3}{1}=\frac{y-k}{2}=\frac{z}{1}$ **intersect then k=**

**(JEE MAINS -2012)**

1) -1 2) 2/9 3) 9/2 4) 0

16. **Let L be the line of intersection of the planes 2x + 3y + z = 1 and x + 3y + 2z = 2. If L makes an angle** $\alpha$ **with the positive X-axis then** $\cos\alpha =$ **(AIEEE - 2007)**

1) $\frac{1}{\sqrt{3}}$ 2) $\frac{1}{2}$ 3) 1 4) $\frac{1}{\sqrt{2}}$

## LEVEL-I (C.W)-KEY

| | | | | | |
|---|---|---|---|---|---|
| 1) 4 | 2) 3 | 3) 3 | 4) 2 | 5) 1 | 6) 3 |
| 7) 3 | 8) 4 | 9) 2 | 10) 1 | 11) 2 | 12) 2 |
| 13) 3 | 14) 2 | 15) 3 | 16) 1 | | |

## LEVEL-I (C.W)-HINTS

1. Use the formula $\frac{x-x_1}{l}=\frac{y-y_1}{m}=\frac{z-z_1}{n}$, where $l=m=n=\frac{1}{\sqrt{3}}$
2. D.r.s of required and given lines are same
3. By verify the options
4. $\frac{x-5}{3}-\frac{y+4}{7}=\frac{z-6}{2}=\lambda$
5. $\frac{x-1}{-3}=\frac{y-2}{\frac{2p}{7}}=\frac{z-3}{2}$ --- (1)

$\frac{x-1}{\frac{-3p}{7}}=\frac{y-5}{1}=\frac{z-6}{-5}$ --- (2)

(1) and (2) are perpendicular.

$a_1a_2+b_1b_2+c_1c_2=0$

6. Use formula,

$$\cos\theta=\frac{|a_1a_2+b_1b_2+c_1c_2|}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$$

7. Take $(a_1,b_1,c_1)=(3,4,5)$ and $(a_2,b_2,c_2)=(2,-2,1)$.
Use

$$\cos\left(\frac{\pi}{2}-\theta\right)=\frac{|a_1a_2+b_1b_2+c_1c_2|}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$$

8. Given lines are

$\frac{x}{3}=\frac{y}{2}=\frac{z}{-6}$ --- (1)

$\frac{x}{2}=\frac{y}{-12}=\frac{z}{-3}$ --- (2)

If $\theta$ is the required angle

$a_1a_2+b_1b_2+c_1c_2=0;\quad \theta=\frac{\pi}{2}$

9. YZ - plane divides the line segment in the ratio $=-x_1:x_2=-5:3$

$\left(0,\frac{5b-3}{2},\frac{5-3a}{2}\right)=\left(0,\frac{17}{2},\frac{-13}{2}\right)$

10. The line x = 1, y = 2 is parallel to z-axis.
The line y = 1, z = 0 is parallel to x-axis.

Angle between the lines is $\frac{\pi}{2}$

11. $\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix}=0$

12. $\frac{x+3}{3}=\frac{y-2}{-2}=\frac{z+1}{1}=t$ ( say),

the point $(3t-3,2-2t,t-1)$ satisfy the given plane

13. $\begin{vmatrix} 1 & -1 & -1 \\ 1 & 1 & -k \\ k & 2 & 1 \end{vmatrix}=0 \Rightarrow k^2+3k=0$, k = 0, –3

14. $(t,t-a,t)$ is a point on first line and $\left(s-a,\frac{s}{2},\frac{s}{2}\right)$ is point on second line,

find d.r.s and compare with given d.r.s (or) verification

15 $\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}=0$

16. D.r's of the line of intersection $\begin{matrix} 3 & & 1 & & 2 & & 3 \\ & \times & & \times & & \times & \\ 3 & & 2 & & 1 & & 3 \end{matrix}$

$(a_1,b_1\ c_1)=(3,-3,3)$

D.r's of x-axis $(a_2,b_2\ c_2)=(1,0,0)$

$\cos\alpha=\frac{3}{\sqrt{9+9+9}}=\frac{1}{\sqrt{3}}$

## LEVEL-I - (H.W)

## EQUATION OF A LINE

**1. The equation of the line joining ( -2,1,3 ) and ( 1,1,4 ) is**

1) $\frac{x+2}{3}=\frac{y-1}{0}=\frac{z-3}{1}$ 2) $\frac{x-2}{3}=\frac{y+1}{0}=\frac{z+3}{1}$

3) $\frac{x+2}{4}=\frac{y+1}{3}=\frac{z-3}{2}$ 4) $\frac{x-3}{1}=\frac{y-1}{1}=\frac{z-2}{1}$

**2. Equation of a line passing through (1,-2,3) and parallel to the plane $2x+3y+z+5=0$ is**

1) $\frac{x-1}{-1}=\frac{y+2}{1}-\frac{z-3}{-1}$ 2) $\frac{x-1}{2}=\frac{y+2}{3}-\frac{z-3}{1}$

3) $\frac{x+1}{-1}=\frac{y-2}{1}-\frac{z-3}{-1}$ 4) $\frac{x+1}{2}=\frac{y-2}{3}-\frac{z-3}{1}$

**3. The equation to the plane which passes through the z-axis and is perpendicular to the line $\frac{x-1}{\cos\alpha}=\frac{y+2}{\sin\alpha}=\frac{z-3}{0}$ is**

1) $x\sin\alpha+y\cos\alpha=0$ 2) $x\sin\alpha-y\cos\alpha=0$

3) $x\cos\alpha+y\sin\alpha=0$ 4) $x\cos\alpha-y\sin\alpha=0$

4. The vector equation of the line $6x-2=3y+1=2z-2$ is

1) $\bar{r}=(\bar{i}-\bar{j}+3\bar{k})+\lambda(\bar{i}+2\bar{j}+3\bar{k})$

2) $\bar{r}=(\bar{i}+2\bar{j}+3\bar{k})+\lambda\left(\frac{1}{3}\bar{i}-\frac{1}{3}\bar{j}+\bar{k}\right)$

3) $\bar{r}=\left(\frac{1}{3}\bar{i}-\frac{1}{3}\bar{j}+\bar{k}\right)+\lambda(\bar{i}+2\bar{j}+3\bar{k})$

4) $\bar{r}=(-2\bar{i}+\bar{j}-2\bar{k})+\lambda(6\bar{i}+3\bar{j}+2\bar{k})$

5. Parametric form of the equation of the line $3x-6y-2z-15=0=2x+y-2z-5$ is

1) $\frac{x-5}{14}=\frac{y}{2}=\frac{z}{15}$ 2) $\frac{x-1}{14}=\frac{y-5}{2}=\frac{z-1}{15}$

3) $\frac{x-3}{14}=\frac{y+1}{2}=\frac{z}{15}$ 4) $\frac{x+5}{14}=\frac{y}{2}=\frac{z}{15}$

## ANGLE BETWEEN TWO LINES

6. The value of k if the lines $\frac{x+1}{-3}=\frac{y+2}{2k}=\frac{z-3}{2}$ and $\frac{x-1}{3k}=\frac{y+5}{1}=\frac{z+6}{7}$ may be perpendicular

1) 3 2) 2 3) -2 4) 4

7. The line $\frac{x-2}{3}=\frac{y-3}{4}=\frac{z-4}{5}$ is perpendicular to the plane

1) $2x+y-2z=0$ 2) $3x+4y+5z=7$

3) $x+y+z=2$ 4) $2x+3y+4z=0$

8. The angle between the pair of lines $\bar{r}=(3\bar{i}+2\bar{j}-4\bar{k})+\lambda(\bar{i}+2\bar{j}+2\bar{k})$ and $\bar{r}=(5\bar{i}-2\bar{j})+\mu(3\bar{i}+2\bar{j}+6\bar{k})$ is

1) $\tan^{-1}(19/21)$ 2) $\cos^{-1}(19/21)$

3) $\sin^{-1}(19/21)$ 4) $\cos^{-1}(19/20)$

9. If $\theta$ is the angle between line $\frac{x+1}{1}=\frac{y-1}{2}=\frac{z-2}{2}$ and the plane $2x-y+\sqrt{\lambda}z+4=0$ is such that $\sin\theta=\frac{1}{3}$ then the value of $\lambda$ is

1) $\frac{5}{3}$ 2) $\frac{-3}{5}$ 3) $\frac{3}{4}$ 4) $\frac{-4}{3}$

10. The angle between the lines $\frac{x-5}{4}=\frac{y-2}{1}=\frac{z-3}{8}$ and $\frac{x}{2}=\frac{y}{2}=\frac{z}{1}$ is

1) $\cos^{-1}(2/3)$ 2) $\cos^{-1}(1/3)$

3) $\cos^{-1}(4/5)$ 4) $\cos^{-1}(4/3)$

11. The angle between the lines $\frac{x-5}{7}=\frac{y+2}{-5}=\frac{z}{1}$ and $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ is

1) $\frac{\pi}{3}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{6}$

## COPLANAR LINES

12. The equation of the plane containing the line $\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-3}{4}$ and perpendicular to the plane $x+2y+z=12$ is

1) $9x-2y+5z+4=0$ 2) $9x-2y-5z+4=0$

3) $9x-2y-5z-4=0$ 4) $9x+2y-5z-4=0$

13. A plane which passes through the point (3, 2, 0) and the line $\frac{x-4}{1}=\frac{y-7}{5}=\frac{z-4}{4}$ is

(AIEEE - 2002)

1) $x-y+z=1$ 2) $x+y+z=5$

3) $x+2y-2=1$ 4) $2x-y+z=5$

14. The value of *m* for which staight line $3x-2y+z+3=0=4x-3y+4z+1$ is parallel to the plane $2x-y+mz-2=0$ is

1) $-2$ 2) 8 3) $-18$ 4) 11

## INTERSECTION OF LINES

15. Let the line $\frac{x-2}{3}=\frac{y-1}{-5}=\frac{z+2}{2}$ lie in the plane $x+3y-\alpha z+\beta=0$ then $(\alpha,\beta)=$

(AIEEE - 2009)

1) $(-6, 7)$ 2) $(5, -15)$ 3) $(-5, 5)$ 4) $(6, -17)$

16. **The d.r's of the line given by the planes $x-y+z-5=0$, $x-3y-6=0$ are**
1) (3, 1,–2) 2) (2, –4, 1)
3) (1, –1, 1) 4) (0, 2, 1)

## LEVEL-I (H.W)-KEY

| | | | | | |
|---|---|---|---|---|---|
| 1) 1 | 2) 1 | 3) 3 | 4) 3 | 5) 3 | 6) 2 |
| 7) 2 | 8) 2 | 9) 1 | 10) 1 | 11) 3 | 12) 2 |
| 13) 1 | 14) 1 | 15) 1 | 16) 1 | | |

## LEVEL-I (H.W)-HINTS

1. D.r's of the line = $(x_2-x_1, y_2-y_1, z_2-z_1)$
Let (a, b, c) = (3, 0, 1)
Use the formula $\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$
2. By verify the options
3. By verification D.r's of normal are $(\cos\alpha, \sin\alpha, 0)$
4. Convert into the form $\frac{x-\frac{1}{3}}{\left(\frac{1}{6}\right)}=\frac{y+\frac{1}{3}}{\left(\frac{1}{3}\right)}+\frac{z-1}{\left(\frac{1}{2}\right)}$
and use $\bar{r}=\bar{a}+t\bar{b}$
5. $\pi_1=3x-6y-2z-15=0$
$\pi_2=2x+y-2z-5=0$
now the dr's of the common line two planes are $(14,2,15)$ and Put $z=0$ *in* $\pi_1$ *and* $\pi_2$ and solve them we get a point
6. $a_1a_2+b_1b_2+c_1c_2=0$
7. By verification method
8. D.r's of given lines
$\vec{b}_1=(1,2,2), \vec{b}_2=(3,2,6);\ \cos\theta=\frac{\vec{b}_1.\vec{b}_2}{|\vec{b}_1||\vec{b}_2|}$
9. Since the line makes an angle $\theta$ with the plane, it makes an angle $\left(\frac{\pi}{2}-\theta\right)$ with the normal to the plane.
$$\therefore\ \cos\left(\frac{\pi}{2}-\theta\right)=\frac{2(1)+(-1)(2)+\left(\sqrt{\lambda}\right)(2)}{\sqrt{1+4+4}\times\sqrt{4+1+\lambda}}$$
$$\Rightarrow\frac{1}{3}=\frac{2\sqrt{\lambda}}{3\sqrt{\lambda+5}}\Rightarrow\lambda+5=4\lambda\ \Rightarrow\lambda=\frac{5}{3}$$
10. Use $\cos\theta$ formula.
11. $a_1a_2+b_1b_2+c_1c_2=0$
12. Take $\begin{vmatrix} x-1 & y+1 & z-3 \\ 2 & -1 & 4 \\ 1 & 2 & 1 \end{vmatrix}=0$
13. Verification
Required plane has to pass through the points (3, 2, 0) and (4, 7, 4)
14. $\pi_1=3x-2y+z+3=0$
$\pi_2=4x-3y+4z+1=0$
$$\begin{array}{cccc} a & b & c & \\ -2 & 1 & 3 & -2 \\ -3 & 4 & 4 & -3 \end{array}\Rightarrow\frac{a}{-8+3}=\frac{b}{4-12}=\frac{c}{-9+8}$$
D.r's of the common line of the two planes are $(-5,-8,-1)$
D.r's of the normal to the plane are (2, –1, m)
Now apply the perpendicular condition with the given plane we get $m=-2$
15. Given line lies in the given plane
$1(3)+3(-5)-\alpha(2)=0 \Rightarrow \alpha=-6$
and $2+3(1)-\alpha(-2)+\beta=0 \Rightarrow \beta=7$
16. By cross multiplication method
$$\begin{array}{cccc} & a & b & c \\ -1 & 1 & 1 & -1 \\ -3 & 0 & 1 & -3 \end{array}\Rightarrow\frac{a}{3}=\frac{b}{1}=\frac{c}{-2}$$

## LEVEL-II - (C.W)

### DISTANCE

1. **The distance of the point (1, -5, 9) from the plane $x-y+z=5$ measured along a straight line $x=y=z$ is (AIEEE-2011)**
1) $3\sqrt{5}$ 2) $10\sqrt{3}$ 3) $5\sqrt{3}$ 4) $3\sqrt{10}$
2. **A line passes through two points A(2,-3,-1) and B(8,-1,2). The coordinates of a point on this line at a distance of 14 units from A are**
1) (14, 1, 5) 2) (-10, -7, 7)
3) (86, 25, 41) 4) $(0,0,0)$

3. **The distance of the point $P(3,8,2)$ from the line $\frac{1}{2}(x-1)=\frac{1}{4}(y-3)=\frac{1}{3}(z-2)$ measured parallel to the plane $3x+2y-2z+15=0$ is**

1) 7 2) 9 3) $\sqrt{7}$ 4) 49

## POINT OF INTERSECTION

4. **The line $\frac{x-2}{3}=\frac{y+1}{2}=\frac{z-1}{-1}$ intersects the curve $xy=c^2, z=0$ if $c=$**

1) $\pm 1$ 2) $\pm\sqrt{3}$ 3) $\pm\sqrt{5}$ 4) $\pm\sqrt{7}$

5. **The point of intersection of the line $\frac{x-3}{3}=\frac{2-y}{4}=\frac{z+1}{1}$ and planes $2x+4y+3z+3=0$, $x+2y+3z=0$ is**

1) ( 9,6,1 ) 2) ( -9,6,1 )
3) ( 9,-6,1 ) 4) ( -9,-6,-1 )

## SHORTEST DISTANCE

6. **The shortest distance between the lines $\frac{x-2}{3}=\frac{y-3}{4}=\frac{z-4}{5}$ ; $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ is**

1) 2 2) 1 3) 3 4) 0

7. **The shortest distance between the lines $\bar{r}=(2\bar{i}-\bar{j}-\bar{k})+\lambda(2\bar{i}+\bar{j}+2\bar{k})$ and $\bar{r}=(\bar{i}+2\bar{j}+\bar{k})+\mu(\bar{i}-\bar{j}+\bar{k})$ is**

1) $3/\sqrt{2}$ 2) $3\sqrt{2}$ 3) $2/\sqrt{3}$ 4) $\sqrt{2}/3$

## DISTANCE BETWEEN PARALLEL LINES

8. **The distance between the parallel lines $\bar{r}=2\bar{i}+3\bar{j}-\bar{k}+\lambda(\bar{i}-\bar{j}+2\bar{k})$ and $\bar{r}=-3\bar{i}+4\bar{j}+\bar{k}+\mu(\bar{i}-\bar{j}+2\bar{k})$ is**

1) $2\sqrt{\frac{22}{3}}$ 2) $\sqrt{\frac{175}{6}}$ 3) $\frac{14}{\sqrt{6}}$ 4) 7

## FOOT AND IMAGE

9. **The reflection of the point A (1, 0, 0) in the line $\frac{x-1}{2}=\frac{y+1}{-3}=\frac{z+10}{8}$ is**

1) (3, –4, –2) 2) (5, –8, –4)
3) (1, –1, –10) 4) (2, –3, 8)

10. **The foot of the perpendicular from (a,b,c) on the line $x=y=z$ is the point $(r,r,r)$ where**

1) $r=a+b+c$ 2) $r=3(a+b+c)$
3) $3r=a+b+c$ 4) $r=4(a+b+c)$

## LENGTH OF THE PERPENDICULAR

11. **The perpendicular distance of the point (2,4,-1) from line $\frac{x+5}{1}=\frac{y+3}{4}=\frac{z-6}{-9}$ is**

1) 3 2) 5 3) 7 4) 9

12. **The length of perpendicular drawn from the point $(3,-1,11)$ to the line $\frac{x}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ is (AIEEE-2011)**

1) $\sqrt{66}$ 2) $\sqrt{29}$ 3) $\sqrt{33}$ 4) $\sqrt{53}$

## LEVEL-II (C.W)-KEY

| | | | | | |
|---|---|---|---|---|---|
| 1) 2 | 2) 1 | 3) 1 | 4) 3 | 5) 3 | 6) 4 |
| 7) 1 | 8) 1 | 9) 2 | 10) 3 | 11) 3 | 12) 4 |

## LEVEL-II (C.W)-HINTS

1. Let P = (1, -5, 9)
Let Q be a point on the given plane such that PQ is parallel to given line
The equation of the line PQ is

$$\frac{x-1}{1}=\frac{y+5}{1}=\frac{z-9}{1}$$

Let Q = $(1+t, t-5, t+9)$
Sub Q in the given plane, $t=-10$
$\therefore Q=(-9,-15,-1)$
PQ = $\sqrt{300}=10\sqrt{3}$

2. $\frac{x-2}{6}=\frac{y+3}{2}=\frac{z+1}{3}\Rightarrow\frac{x-2}{\frac{6}{7}}=\frac{y+3}{\frac{2}{7}}=\frac{z+1}{3/7}$ points on the line

$=\left(2\pm\frac{6}{7}r,-3\pm\frac{2r}{7},-1\pm\frac{3r}{7}\right)$ where $r$=14.

3. Let $A=(2\lambda+1,4\lambda+3,3\lambda+2)$

Let this point lie on the line such that AP is parallel to the plane

$\overline{AP}$ is perpendicular to normal to the plane.

$\Rightarrow \overline{AP}\perp\left(3\hat{i}+2\hat{j}-2\hat{k}\right)$

$\Rightarrow 3.(2\lambda-2)+2(4\lambda-5)-2(3\lambda)=0$

$\Rightarrow \lambda=2$

therefore, A is (5,11,8), P = (3, 8, 2) , AP = 7

4. we have $z=0$ for the point, where the line intersects the curve. therefore,

$\frac{x-2}{3}=\frac{y+1}{2}=\frac{0-1}{-1}$

$\Rightarrow \frac{x-2}{3}=1$ and $\frac{y+1}{2}=1$

$\Rightarrow x=5$ and $y=1$

Putting these values in $xy=c^2$, we get

$5=c^2\Rightarrow c=\pm\sqrt{5}$

5. By verification method

6. $\bar{a}_1=(2,3,4)$, $\bar{a}_2=(1,2,3)$, $\bar{b}_1=(3,4,5)$

$\bar{b}_2=(2,3,4)$ , $\left[\bar{a}_1-\bar{a}_2\ \ \bar{b}_1\ \ \bar{b}_2\right]=0$

7. $\bar{a}_1=(2,-1,-1)$, $\bar{a}_2=(1,2,1)$, $\bar{b}_1=(2,1,2)$,

$\bar{b}_2=(1,-1,1)$, $\bar{a}_1-\bar{a}_2=(1,-3,-2)$

$\left[\bar{a}_1-\bar{a}_2\ \ \bar{b}_1\ \ \bar{b}_2\right]=9$ , $\bar{b}_1\times\bar{b}_2=3\bar{i}+3\bar{k}$

$\left|\bar{b}_1\times\bar{b}_2\right|=3\sqrt{2}$

8. $\bar{a}_1=2\bar{i}+3\bar{j}-\bar{k}$, $\bar{a}_2=-3\bar{i}+4\bar{j}+\bar{k}$,

$\bar{b}=\bar{i}-\bar{j}+2\bar{k}$ , $\bar{a}_1-\bar{a}_2=5\bar{i}-\bar{j}-2\bar{k}$

$$\bar{b}\times(\bar{a}_1-\bar{a}_2)=\begin{vmatrix}\bar{i}&\bar{j}&\bar{k}\\1&-1&2\\5&-1&-2\end{vmatrix}=4\bar{i}+12\bar{j}+4\bar{k}$$

$\left|\bar{b}\times(\bar{a}_1-\bar{a}_2)\right|=\sqrt{176}$ , $\left|\bar{b}\right|=\sqrt{6}$

Distance $=\frac{\left|\bar{b}\times(\bar{a}_1-\bar{a}_2)\right|}{|\bar{b}|}=\frac{\sqrt{176}}{\sqrt{6}}=2\sqrt{\frac{22}{3}}$

9. Any point P on the line is (2r+1, –3r–1, 8r–10)

D.r's of AP are (2r, –3r–1, 8r–10)

AP is perpendicular to the given line

2(2r) – 3(–3r–1) + 8(8r–10) = 0 $\Rightarrow$ r = 1

P (3, –4, –2)

Let B be the image of A

B = 2P – A

10. apply the formula for perpendicular condition.

11. $A=(2,4,-1)$,

Any point on the line $P=(t-5,4t-3,-9t+6)$

Let P be the foot of the perpendicular of A

AP is perpendicular to given line.

Use formula $a_1a_2+b_1b_2+c_1c_2=0$ , t = 1

12. Let $p=(3,-1,11)$

Let Q be the foot of the perpendicular of P on the given line

Let the coordinates of Q be

$(2t,3t+2,4t+3)$ which is any point on $\overline{AB}$

D.r's of PQ = $(2t-3,3t+3,4t-8)$

D'r's of AB = (2, 3, 4)

AB is perpendicular to PQ

$2(2t-3)+3(3t+3)+4(4t-8)=0$

$\Rightarrow t=1,\qquad Q=(2,5,7)$, PQ = $\sqrt{53}$

## LEVEL-II - (H.W)

### DISTANCE

**1. The distance of the point (3, -4, 5) from the plane $2x+5y-6z=16$ measured along a line with d.cs, proportional to (2, 1, -2) is**

1) $\frac{60}{7}$ 2) $\frac{50}{7}$ 3) $\frac{30}{7}$ 4) $\frac{20}{7}$

2. The distance of the point (1,0,–3) from the plane $x-y-z=9$ measured parallel to the line $\frac{x-2}{2}=\frac{y+2}{3}=\frac{z-6}{-6}$ is

1) 6 2) 7 3) 8 4) 9

## POINT OF INTERSECTION

3. A line with d.r's (2,7,–5) is drawn to intersect the lines $\frac{x-5}{3}=\frac{y-7}{-1}=\frac{z+2}{1}$ and $\frac{x+3}{-3}=\frac{y-3}{2}=\frac{z-6}{4}$ at P and Q respectively. Length of PQ is

1) $\sqrt{78}$ 2) $\sqrt{77}$ 3) $\sqrt{54}$ 4) $\sqrt{74}$

4. If the straight lines $\frac{x-1}{k}=\frac{y-2}{2}=\frac{z-3}{3}$ and $\frac{x-2}{3}=\frac{y-3}{k}=\frac{z-1}{2}$ intersect at a point then the integer k is equal to

1) 5 2) 2 3) –2 4) –5

## SHORTEST DISTANCE

5. The shortest distance between the lines $\frac{x-2}{3}=\frac{y-3}{4}=\frac{z-1}{2};\frac{x-4}{4}=\frac{y-5}{5}=\frac{z-2}{3}$ is

1) $\frac{1}{\sqrt{3}}$ 2) $\frac{1}{\sqrt{6}}$ 3) $\frac{1}{\sqrt{2}}$ 4) $\frac{5}{\sqrt{6}}$

6. The shortest distance between the lines $\bar{r}=(1-t)\bar{i}+(t-2)\bar{j}+(3-2t)\bar{k}$ and $\bar{r}=(s+1)\bar{i}+(2s-1)\bar{j}-(2s+1)\bar{k}$ is

1) $8/\sqrt{17}$ 2) $8/\sqrt{493}$ 3) $8/\sqrt{29}$ 4) $16\sqrt{29}$

7. The length of the shortest distance between the lines $\frac{x-3}{1}=\frac{y-5}{-2}=\frac{z-7}{1}$ and $\frac{x+1}{7}=\frac{y+1}{-6}=\frac{z+1}{1}$ is

1) $4\sqrt{29}$ 2) $3\sqrt{29}$ 3) $\sqrt{29}$ 4) $2\sqrt{29}$

## DISTANCE BETWEEN PARALLEL LINES

8. The distance between the parallel lines $\bar{r}=(\bar{i}-\bar{j})+\lambda(3\bar{i}+\bar{j}-\bar{k})$ and $\bar{r}=(\bar{j}-\bar{k})+\mu(2\bar{i}+\bar{j}-\bar{k})$ is

1) 6 2) 7 3) $\sqrt{\frac{35}{6}}$ 4) $\sqrt{\frac{33}{6}}$

## FOOT AND IMAGE

9. Foot of the perpendicular drawn from the point A (1,0,3) to the join of the points B (4,7,1) and C (3,5,3) is

1) (1,2-2) 2) $\left(\frac{5}{3},\frac{7}{3},\frac{17}{3}\right)$

3) $\left(\frac{1}{3},\frac{2}{3},\frac{-2}{3}\right)$ 4) $\left(\frac{-5}{3},\frac{-7}{3},\frac{-17}{3}\right)$

10. The image of the point (3, –1, 11) w.r.t the line $\frac{x}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ is

1) (2, 5, 7) 2) (1, 11, 3)
3) (0, 0, 0) 4) (0,2,3)

## LENGTH OF THE ERPENDICULAR

11. The length of the perpendicular from the point (1, 2, 3) to the line $\frac{x-6}{3}=\frac{y-7}{2}=\frac{z-7}{-2}$ is

1) 7 2) $\sqrt{48}$ 3) 8 4) 9

12. The length of the perpendicular from the point (–1, 3, 9) to the line $\frac{x-13}{5}=\frac{y+8}{-8}=\frac{z-31}{1}$ is

1) 21 2) 22 3) 20 4) $\sqrt{439}$

## LEVEL-II (H.W)-KEY

| | | | | | |
|---|---|---|---|---|---|
| 1) 1 | 2) 2 | 3) 1 | 4) 4 | 5) 2 | 6) 3 |
| 7) 4 | 8) 3 | 9) 2 | 10) 2 | 11) 1 | 12) 1 |

## LEVEL-II (H.W)-HINTS

1. Equation of the line passing through A (3,–4, 5) with d.r's (2, 1,–2) is $\frac{x-3}{2}=\frac{y+4}{1}=\frac{z-5}{-2}=t$

x = 3 + 2t, y = –4 + t, z = 5 – 2t

Let the line passing through A and parallel to given line intersects the plane at P

Let P = (3+2t, –4+t, 5–2t)

Substitute P in the given plane, $t = \frac{20}{7}$.

$$P = \left(\frac{61}{7}, \frac{-8}{7}, \frac{-5}{7}\right); \quad AP = \frac{60}{7}$$

2. Equation of the line passing through (1, 0, –3) with d.r's (2, 3, –6) is

$$\frac{x-1}{2} = \frac{y-0}{3} = \frac{z+3}{-6} = t \text{ (say)}$$

x = 1 + 2t, y = 3t, z = –3 – 6t

Let P be a point in the plane x–y–z=9 such that AP is parallel to given line

P = (1 + 2t, 3t, –3 – 6t)

Substitute P in the given plane, t = 1

P = (3, 3, –9), AP = 7.

3. P = (5+3t, 7–t, –2+t),

Q = (–3–3$\lambda$, 3+2$\lambda$, 6+4$\lambda$)

D.r's of PQ = (8 + 3t + 3$\lambda$, 4–t– 2$\lambda$, t–4$\lambda$–8)

= (2,7,–5) $\Rightarrow$ t = $\lambda$ = –1,

P = (2, 8, –3), Q = (0, 1, 2)

PQ = $\sqrt{78}$

4. $$\begin{vmatrix} 1 & 1 & -2 \\ k & 2 & 3 \\ 3 & k & 2 \end{vmatrix} = 0 \Rightarrow k = -5$$

5. $\bar{a}_1 = (2,3,1)$, $\bar{a}_2 = (4,5,2)$, $\bar{b}_1 = (3,4,2)$

$\bar{b}_2 = (4,5,3)$

$$\text{Shortest distance} = \frac{\left|\left[\bar{a}_1 - \bar{a}_2 \ \ \bar{b}_1 \ \ \bar{b}_2\right]\right|}{\left|\bar{b}_1 \times \bar{b}_2\right|}$$

6. Given lines

$\bar{r} = (\bar{i} - 2\bar{j} + 3\bar{k}) + t(-\bar{i} + \bar{j} - 2\bar{k})$ and

$\bar{r} = (\bar{i} - \bar{j} - \bar{k}) + s(\bar{i} + 2\bar{j} - 2\bar{k})$

$\vec{a}_1 = (1,-2,3), \vec{b}_1 = (-1,1,-2)$

$\vec{a}_2 = (1,-1,-1), \vec{b}_2 = (1,2,-2)$

$$\text{Find } \frac{\left|\left[\bar{a}_1 - \bar{a}_2 \ \ \bar{b}_1 \ \ \bar{b}_2\right]\right|}{\left|\bar{b}_1 \times \bar{b}_2\right|}$$

7. $\bar{a}_1 = (3,5,7)$, $\bar{a}_2 = (-1,-1,-1)$

$\bar{a}_1 - \bar{a}_2 = (4,6,8)$

$\bar{b}_1 = (1,-2,1), \bar{b}_2 = (7,-6,1)$

$\left[\bar{a}_1 - \bar{a}_2 \ \ \bar{b}_1 \ \ \bar{b}_2\right] = 116$

$\left|\bar{b}_1 \times \bar{b}_2\right| = \sqrt{116}$

$$\frac{\left|\left[\bar{a}_1 - \bar{a}_2 \ \ \bar{b}_1 \ \ \bar{b}_2\right]\right|}{\left|\bar{b}_1 \times \bar{b}_2\right|} = \frac{116}{\sqrt{116}} = 2\sqrt{29}$$

8. $\bar{a}_1 = \bar{i} - \bar{j}$, $\bar{a}_2 = \bar{j} - \bar{k}$, $\bar{b} = 2\bar{i} + \bar{j} - \bar{k}$

$\bar{a}_1 - \bar{a}_2 = \bar{i} - 2\bar{j} + \bar{k}$

$$\bar{b} \times (\bar{a}_1 - \bar{a}_2) = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 2 & 1 & -1 \\ 1 & -2 & 1 \end{vmatrix} = -\bar{i} - 3\bar{j} - 5\bar{k}$$

$\left|\bar{b} \times (\bar{a}_1 - \bar{a}_2)\right| = \sqrt{35}$

$\left|\bar{b}\right| = \sqrt{6}$, Distance = $\sqrt{\frac{35}{6}}$

9. A = (1, 0, 3)

D.r's of the line BC = (1, 2, –2)

Equation of the line BC is $\frac{x-4}{1} = \frac{y-7}{2} = \frac{z-1}{-2}$

Any point P on the line BC is (4+t, 7+2t, 1–2t)

D.r's of P = (3 + t, 7 + 2t, –2t – 2)

AP is perpendicular to BC

1(3 + t) + 2(7 + 2t) + 2(2t + 2) = 0

$$t = \frac{-7}{3}, P = \left(4 - \frac{7}{3}, 7 - \frac{14}{3}, 1 + \frac{14}{3}\right)$$

$$P = \left(\frac{5}{3}, \frac{7}{3}, \frac{17}{3}\right)$$

10. Let P be the foot of the perpendicular from A (3, –1, 11) to the given line then

P = (2r, 3r+2, 4r+3)

D.r's of $\overrightarrow{AP}$ are (2r – 3, 3r + 3, 4r – 8)

$\overrightarrow{AP}$ is perpendicular to the given line

$\Rightarrow 2(2r-3)+3(3r+3)+4(4r-8)=0 \Rightarrow r=1$

P = (2, 5, 7)

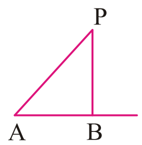
Image = 2P – A = (1, 11, 3)

11. Let P = (1, 2, 3)

Let A = (6, 7, 7)

Let B be the foot of the perpendicular of P on the given line

D.r's of given line = (3, 2, –2)




D.c's of given line $=\left(\frac{3}{\sqrt{17}},\frac{2}{\sqrt{17}},\frac{-2}{\sqrt{17}}\right)$

AB = projection of AP on the given line

$=\left|l(x_2-x_1)+m(y_2-y_1)+n(z_2-z_1)\right|$

AB = $\sqrt{17}$

From $\Delta$ABP, $PB^2=AP^2-AB^2=66-17$

PB = $\sqrt{49}=7$

12. Let A = (–1, 3, 9)

Any point P on the line is

(13 + 5t, –8–8t, 31 + t)

Let P be the foot of the perpendicular of A

D.r's of AP = (14+5t, –11–8t, 22 + t)

AP is perpendicular to given line.

$a_1a_2+b_1b_2+c_1c_2=0$

$\Rightarrow 5(14+5t)+8(11+8t)+22+t=0$

$\Rightarrow t=-2$ , P = (3, 8, 29), AP = 21

## LEVEL - III

**1. A plane mirror is placed at the origin so that the direction ratios of its normal are (1, -1, 1). A ray of light, coming along the positive direction of the *x*-axis strikes the mirror. Then the direction ratios of the reflected ray are**

1) $\frac{1}{3},\frac{2}{3},\frac{2}{3}$ 2) $\frac{-1}{3},\frac{2}{3},\frac{2}{3}$

3) $\frac{-1}{3},\frac{-2}{3},\frac{-2}{3}$ 4) $\frac{-1}{3},\frac{-2}{3},\frac{2}{3}$

**2. Equation of the perpendicular line from (3, –1, 11) to the line $\frac{x}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ is**

1) $\frac{x-3}{1}=\frac{y+1}{-6}=\frac{z-11}{4}$ 2) $\frac{x-3}{2}=\frac{y+1}{5}=\frac{z-11}{7}$

3) $\frac{x-3}{1}=\frac{y+1}{11}=\frac{z-11}{3}$ 4) $\frac{x-3}{1}=\frac{y+1}{6}=\frac{z-11}{4}$

**3. The equation of line of shortest distance between the lines $\frac{x-3}{1}=\frac{y-5}{-2}=\frac{z-7}{1}$ ; $\frac{x+1}{7}=\frac{y+1}{-6}=\frac{z+1}{1}$ is**

1) $\frac{x-3}{2}=\frac{y-5}{3}=\frac{z-7}{4}$ 2) $\frac{x+3}{2}=\frac{y+5}{3}=\frac{z-7}{4}$

3) $\frac{x+3}{2}=\frac{y-5}{3}=\frac{z-7}{4}$ 4) $\frac{x-3}{2}=\frac{y-5}{3}=\frac{z+7}{4}$

**4. Let $P(3,2,6)$ be a point in space and Q be a point on the line $\bar{r}=(\bar{i}-\bar{j}+2\bar{k})+\mu(-3\bar{i}+\bar{j}+5\bar{k})$ then the value of $\mu$ for which the vector $\overline{PQ}$ is parallel to the plane $x-4y+3z=1$ is (IIT-2009)**

1) $\frac{1}{4}$ 2) $\frac{-1}{4}$ 3) $\frac{1}{8}$ 4) $\frac{-1}{8}$

5. **If the angle between the line $x=\frac{y-1}{2}=\frac{z-3}{\lambda}$ and the plane $x+2y+3z=4$ is $\cos^{-1}\left(\sqrt{5/14}\right)$ then $\lambda=$(AIEEE-2011)**

1) $\frac{3}{2}$ 2) $\frac{5}{3}$ 3) $\frac{2}{3}$ 4) $\frac{2}{5}$

6. **If lines $x=y=z$ and $x=y/2=z/3$ and third line passing through $(1,1,1)$ form a triangle of area $\sqrt{6}$ units, then point of intersection of third line with second line will be**

1) $(1,2,3)$ 2) $(2,4,6)$

3) $\left(\frac{4}{3},\frac{8}{3},\frac{12}{3}\right)$ 4) $(2,1,3)$

7. **The image of the line $\frac{x-1}{3}=\frac{y-3}{1}=\frac{z-4}{-5}$ in the plane 2x – y + z + 3 = 0 is the line (MAINS-2014)**

1) $\frac{x-3}{3}=\frac{y+5}{1}=\frac{z-2}{-5}$ 2) $\frac{x-3}{-3}=\frac{y+5}{-1}=\frac{z-2}{5}$

3) $\frac{x+3}{3}=\frac{y-5}{1}=\frac{z-2}{-5}$ 4) $\frac{x+3}{-3}=\frac{y-5}{-1}=\frac{z+2}{5}$

8. **For the line $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{3}$, which one of the following is incorrect ?**

1) it lies in the plane $x-2y+z=0$

2) it is same as line $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$

3) it passes through (2,3,5)

4) it is parallel to the plane $x-2y+z=6$

9. The equation of a plane which passes through the point of intersection of lines $\frac{x-1}{3}=\frac{y-2}{1}=\frac{z-3}{2}$ and $\frac{x-3}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ and at greatest distance from point (0,0,0) is

1) $4x+3y+5z=25$ 2) $4x+3y+5z=50$

3) $4x+3y+5z=49$ 4) $x+7y-5z=2$

10. **The equation of the plane which passes through the z-axis and is perpendcular to the line $\frac{x-a}{\cos\theta}=\frac{y+2}{\sin\theta}=\frac{z-3}{0}$ is**

1) $x+y\tan\theta=0$ 2) $y+x\tan\theta=0$

3) $x\cos\theta-y\sin\theta=0$ 4) $x\sin\theta-y\cos\theta=0$

11. **If the straight lines x = 1 + s, y = –3 – $\lambda$ s, z = 1 + $\lambda$ s and $x=\frac{t}{2},y=1+t,z=2-t$ with parameters 's' and 't' respectively are coplanar then $\lambda$ =**

1) –2 2) 0 3) $-\frac{1}{2}$ 4) –1

12. **The projection of the line $\frac{x+1}{-1}=\frac{y}{2}=\frac{z-1}{3}$ in the plane x-2y+z=6 is the line of intersection of this plane with the plane is**

1) 2x+y+2=0 2) 3x+y-z=2

3) 2x-3y+8z=3 4) x+y-z=1

## LEVEL-III-KEY

| | | | | | |
|---|---|---|---|---|---|
| 1) 4 | 2) 1 | 3) 1 | 4) 1 | 5) 3 | 6) 2 |
| 7) 3 | 8) 3 | 9) 2 | 10) 1 | 11) 1 | 12) 1 |

## LEVEL-III-HINTS

1. Let (*l, m, n*) be the d.c's of reflected ray

We have (1, 0, 0) are the d.c's of incident ray (x-axis).

D.c's of normal are $\left(\frac{1}{\sqrt{3}},-\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$

Let $(l_1,m_1,n_1)=(l,m,n),\quad (l_2,m_2,n_2)=(1,0,0)$

If $\theta$ is the angle between the normal to the plane and incident ray, then $\cos\theta=\frac{1}{\sqrt{3}}$

$$\left(\frac{l_1+l_2}{2\cos\theta},\frac{m_1+m_2}{2\cos\theta},\frac{n_1+n_2}{2\cos\theta}\right)=(l,m,n)$$

$$\frac{l+1}{2\cos\theta}=\frac{1}{\sqrt{3}}\quad\Rightarrow\quad l=\frac{-1}{3}$$

$$\frac{m-0}{2\cos\theta}=\frac{-1}{\sqrt{3}}\Rightarrow m=\frac{-2}{3}$$

$\frac{n+0}{2\cos\theta}=\frac{1}{\sqrt{3}} \Rightarrow n=\frac{2}{3}$

2. Let P be the foot of the perpendicular from A (3, –1, 11) to the given line then

P = (2r , 3r + 2, 4r + 3)

D.r's of A.P are (2r – 3, 3r + 3, 4r – 8)

$\overleftrightarrow{AP}$ is perpendiciular to the given line

$\Rightarrow 2(2r-3)+3(3r+3)+4(4r-8)=0 \Rightarrow r=1$

P = (2, 5, 7)

D.r's of $\overleftrightarrow{AP}$ are (1, –6, 4)

3. Let $P(\alpha+3,-2\alpha+5,\alpha+7)$ and $Q(7\beta-1,-6\beta-1,\beta-1)$ be the points on the given lines so that PQ is the line of shortest distance between the given lines

D.r's of $PQ=(\alpha-7\beta+4,-2\alpha+6\beta+6,\alpha-\beta+8)$

Since PQ is perpendicular to the given lines

$1(\alpha-7\beta+4)-2(-2\alpha+6\beta+6)+1(\alpha-\beta+8)=0$

$\Rightarrow 6\alpha-20\beta=0 \Rightarrow 3\alpha-10\beta=0$ --- (1)

and,

$\Rightarrow 7(\alpha-7\beta+4)-6(-2\alpha+6\beta+6)+1(\alpha-\beta+8)=0$

$\Rightarrow 7\alpha-49\beta+28+12\alpha-36\beta-36+\alpha-\beta+8=0$

$\Rightarrow 20\alpha-86\beta=0 \Rightarrow 10\alpha=43\beta$ --- (2)

From (1) and (2) $\alpha=0,\beta=0$

$P=(3,5,7),\ Q=(-1,-1,-1)$

D.r's of $PQ$ = (–4, –6, –8) = (2, 3, 4)

Equation of PQ is $\frac{x-3}{2}=\frac{y-5}{3}=\frac{z-7}{4}$

4. $P=(3,2,6)$

$Q=(1-3\mu,\mu-1,2+5\mu)$

D.r's of PQ = $(-3\mu-2,\mu-3,5\mu-4)$

Equation of the plane x – 4y + 3z = 1

D.r's of the normal to the plane (1, –4, 3)

PQ is perpendicular to the normal to the plane

$a_1a_2+b_1b_2+c_1c_2=0$

$\Rightarrow -3\mu-2-4\mu+12+15\mu-12=0$

$\Rightarrow 8\mu-2=0 \Rightarrow \mu=\frac{1}{4}$

5. Let $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right)=\theta \Rightarrow \cos\theta=\frac{\sqrt{5}}{\sqrt{14}}$

$\Rightarrow \sin\theta=\frac{3}{\sqrt{14}}$

D.r's of given line $(a_1, b_1, c_1) = (1, 2, \lambda)$

D.r's of normal to the given plane

$(a_2, b_2, c_2) = (1, 2, 3)$

$\sin\theta=\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$

$\frac{3}{\sqrt{14}}=\frac{1+4+3\lambda}{\sqrt{1+4+\lambda^2}\sqrt{1+4+9}};\quad \lambda=\frac{2}{3}$

6. x = y = z--- (1), $x=\frac{y}{2}=\frac{z}{3}$ --- (2)

Clearly point of intersection of (1) and (2) is (0,0,0)

D.r's of (1) are (1, 1, 1)

D.r's of (2) are (1, 2, 3)

Let $\theta$ be the angle between (1) and (2)

$\cos\theta=\frac{6}{\sqrt{42}},\sin\theta=\frac{\sqrt{6}}{\sqrt{42}}$

Let any point on second line be $(\lambda,2\lambda,3\lambda)$

Third line passing through (1, 1, 1)

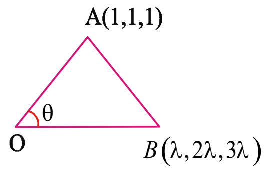
(1, 1, 1) lies on (1)

A = (1, 1, 1)

$Area\ of\ \Delta OAB = \frac{1}{2}(OA)OB\sin\theta$

$= \frac{1}{2}\sqrt{3}\lambda\sqrt{14} \times \frac{\sqrt{6}}{\sqrt{42}} = \sqrt{6} \quad \Rightarrow \lambda = 2$

So B is $(2,4,6)$




7. $3(2) + 1(-1) + (-5)(1) = 0$

Given line and given plane are parallel

$\therefore$ Image line is also parallel to the given line

Image of A (1, 3, 4) w.r.to given plane lies on the image line.

Equation of the normal to the plane is $\frac{x-1}{2} = \frac{y-3}{-1} = \frac{z-4}{1}$

Any point on the line B = (2r + 1, –r+3, r+4)

If B is the image of A (1, 3, 4) then mid point of AB lies on the plane.

Mid point $= \left(\frac{2r+2}{2}, \frac{-r+6}{2}, \frac{r+8}{2}\right)$

Mid point lies in the given plane

$\Rightarrow 2\left(\frac{2r+2}{2}\right) - \left(\frac{-r+6}{2}\right) + \frac{r+8}{2} + 3 = 0$

$\Rightarrow r = -2$ , B = (–3, 5, 2)

Image line is $\frac{x+3}{3} = \frac{y-5}{1} = \frac{z-2}{-5}$

8. (1,2,3) satisfies the plane $x - 2y + z = 0$ and also

$(\hat{i} + 2j + 3k).(i - 2j + k) = 0$

Since the lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ both satisfy (0,0,0) and (1,2,3) both are same. given line is obviously parallel to the plane $x - 2y + z = 6$

9. Let a point $(3\lambda + 1, \lambda + 2, 2\lambda + 3)$ of the first line also lies on the second line Then

$\frac{3\lambda + 1 - 3}{1} = \frac{\lambda + 2 - 1}{2} = \frac{2\lambda + 3 - 2}{3} \Rightarrow \lambda = 1$

Hence, the point of intersection P of the two lines is (4,3,5)

10. The d.r's of the normal of the plane are $(\cos\theta, \sin\theta, 0)$.

Now, the required plane passes through the z-axis. hence the point (0,0,0) lies on the plane.

The required plane is

$x\cos\theta + y\sin\theta = 0, \quad \Rightarrow x + y\tan\theta = 0$

11. Given lines are $\frac{x-1}{1} = \frac{y+3}{-\lambda} = \frac{z-1}{\lambda}$,

$\frac{x}{\frac{1}{2}} = \frac{y-1}{1} = \frac{z-2}{-1}$

Given lines are coplanar

$\Rightarrow \begin{vmatrix} 1 & -4 & -1 \\ 1 & -\lambda & \lambda \\ \frac{1}{2} & 1 & -1 \end{vmatrix} = 0, \quad \Rightarrow \lambda = -2$

12. Equation of a plane through (-1,0,1) is $a(x+1) + b(y-0) + c(z-1) = 0$

Which is paraallel to the given line and perpendicular to the given plane.

$-a + 2b + 3c = 0$ and $a - 2b + c = 0$

by solving the above, we get , c=0, a=2b

## LEVEL - IV

**ASSERTION & REASON:**

**The following questions consist of two statements *one labelled as , Assertion (A) and the other labelled as Reason (R) Y ou are to examine these two statements carefully and decide if the Assertion (A) and the Reason (R) are individually true and if so,whether the Reason (R) is the correct explanation for the given Assertion (A) select your answer to these items using the codes given below and then select the correct option***

***Codes:***

***(A)** Both A and R are individually true and R is the correct explanation of A*

***(B)** Both A and R individually true but R is not the correct explanation of A*

***(C)** A is true but R is false*

***(D)** A is false but R is true*

**1.** **Assertion (A) :The angle between the lines**

$\frac{x}{1}=\frac{y}{1}=\frac{z}{0}$ *and* $\frac{x-1}{2}=\frac{y-2}{1}=\frac{z+3}{2}$ *is* $\frac{\pi}{4}$

**Reason (R): The acute angle between the lines given by** $\frac{x-x_1}{l}=\frac{y-y_1}{m}=\frac{z-z_1}{n}$ **and** $\frac{x-x_2}{p}=\frac{y-y_2}{q}=\frac{z-z_2}{r}$ **then**

$$\cos\theta=\left|\frac{pl+qm+rn}{\sqrt{\sum l^2}\sqrt{\sum p^2}}\right|$$

1) A 2) B 3) C 4) D

**2.** **Statement-1: The point** $A(1,0,7)$ **is the mirror image of the point** $B(1,6,3)$ **in the line** $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$

**Statement-2: The line** $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ **bisects the line segment joining** $A(1,0,7)$ **and** $B(1,6,3)$ **[AIEEE-2011]**

1) Statement-1 is true, statement -2 is true; statement - 2 is not correct explanation for statement-1

2) Statement-1 is true, statement-2 is false

3) Statement -1 is false, statement-2 is true

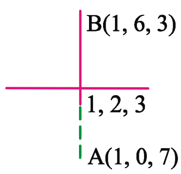
4) Statement-1 is true, statement -2 is true, statement -2 is a correct explanation for statement-1.

## LEVEL-IV-KEY

1) 1 2) 1

## LEVEL-IV-HINTS

1. By using formula

2. 


Statement-1: AB is perpendicular to given line and midpoint of AB lines on line

Statement-2 is true but it is not correct explanation as it is bisector only.

If it is perpendicular bisector then only statement-2 is correct explanation.

**Statement-1**: AB is perpendicular to given line and midpoint of AB lines on line

Statement-2 is true but it is not correct explanation as it is bisector only.

If it is perpendicular bisector then only statement-2 is correct explanation.

$\cos\theta(x-0)+\sin\theta(y-0)+0(z-0)=0$

**Statement-2** is not correct explanation for statement-1